

CHAO SIVCHHUONG

Fifth-Year Mechanical Engineering Student | Institute of Technology of Cambodia

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LinkedIn: [Chao Sivchhuong | LinkedIn](#) **Portfolio:** <https://chaosivchhuong.figma.site>

SUMMARY

Fifth-year Mechanical Engineering student at the Institute of Technology of Cambodia with interests in thermal-fluid systems, computational engineering, materials, and engineering simulation. Experienced in academic research, numerical modeling, CAD, CFD/FEA, and engineering projects using COMSOL Multiphysics, ANSYS, Abaqus, MATLAB, and Python. Recognized through Dean's List honors and competitive scholarships, with experience in both research and industrial engineering. Seeking graduate study to develop advanced engineering and research capabilities.

EDUCATION

Mongkol Borey High School

High School Diploma, Grade A: 2021 – 2022

Institute of Technology of Cambodia

B.Eng. in Mechanical Engineering: 2022 – Present

ACADEMIC HONORS & SCHOLARSHIPS

- Dean's List Honors: 2022 – Present
- Lotte Scholarship Foundation: 2024 – 2025
- Sumitomo Corporation Scholarship: 2023 – 2024
- High School Diploma, Grade A: 2021 – 2022

RESEARCH EXPERIENCE

Steel Furnace Thermal & Fluid Simulation at ITC

Jul 2026 – Oct 2026

- Studying about the charcoal's properties like porosity, permeability
- Building the sample machine to measure the pressure drop through the charcoal's packed bed
- Running the simulation to test the performance of machine and effective of air inlet

PROFESSIONAL EXPERIENCE

Maintenance Engineering Intern at Sailun Tire Industry

Jul 2025 – Oct 2025

- Supported maintenance activities for 7 industrial machines.
- Assisted with machine inspection, troubleshooting, and maintenance tasks.
- Created 2D drawings and 3D models of mechanical components.
- Gained practical experience in tire manufacturing and industrial maintenance.

SELECTED RESEARCH & ENGINEERING PROJECTS

- **Steel Furnace Thermal & Fluid Simulation:** Using COMSOL, Poros Media, Charcoal's and iron ore's properties
- **Finite Element Analysis:** Structural Analysis, Finite Element Methods, Computational Mechanics
- **Supersonic Flight Analysis:** Aerodynamics, Fluid Mechanics, Computational Analysis
- **AI-Based Object Detection:** Python, YOLOv8, Computer Vision
- **Small-Scale Wind Turbine Design:** Mechanical Design, Aerodynamics, Renewable Energy
- **STEM Outreach STEAMING Cambodia (Siemens Green Power Car):** Renewable Energy, STEM Education, Technical Outreach

TECHNICAL SKILLS

- **CAD:** AutoCAD, SolidWorks, Siemens NX, Solid Edge
- **CAE/Simulation:** ANSYS, Abaqus, COMSOL Multiphysics
- **Programming & Analysis:** MATLAB, Python, Excel
- **Engineering:** Mechanical Design, CFD, FEA, Material Testing, Maintenance Engineering
- **Professional Skills:** Research & Analysis, Problem Solving, Teamwork, Leadership, Communication
- **Research & Analysis**
- **Problem Solving**

LANGUAGES

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|------------------|-----------------|
| • Khmer: Native | • French: B1 |
| • English: B1-B2 | • Chinese: HSK2 |

CERTIFICATIONS

CERTIFICATE

- Solid Edge Associate Certification
- Solid Edge Professional Certification
- MATLAB Certification

TRAINING

- BC Course at IFL

My Portfolio



